Designing a Reward / Cost Function for Multi-Objective Unitary Synthesis in `gqe-torch`

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1 Purpose

This document is a *design proposal* for a reward / cost function suitable for the current `gqe-torch` training setup. The goal is to produce a scalar objective that:

- strongly prioritizes process fidelity early in training;
- increasingly favors low depth and low CNOT count once fidelity is already good;

- preserves an exponential / logarithmic coupling between fidelity and structure;
- avoids numerical pathologies at $F = 0$ and $F = 1$;
- remains compatible with replay-based GRPO training.

The core problem is multi-objective:

- maximize fidelity F;
- minimize circuit depth D;
- minimize CNOT count C.

In the current codebase, total gate count is fixed by construction, so it is not a useful training objective yet.

2 Project Constraints

Any reward / cost used here should respect the following properties of the current system:

1. The generator emits *fixed-length* token sequences.
2. The replay buffer stores scalar costs, and those costs are reused across later updates.
3. The training loop is not using a stepwise environment reward; it is scoring complete circuits.
4. Reference quantities that move too quickly can make replay targets inconsistent.
5. Small differences near $F \approx 1$ matter much more than similar absolute differences near $F \approx 0.7$.

These constraints mean the training objective should be as close to *stationary* as possible. For that reason, a pure "reward = change from previous epoch" formulation is not ideal for replay training, even if it is intuitive for human interpretation.

3 Why the Naive Exponential Proposal Is Risky

The intuitive form

$$r = \left(\alpha \frac{D^* - 1}{D_{\text{current}}} + \beta \frac{C^* - 1}{C_{\text{current}}} \right)^{\frac{1}{1-F}}$$

captures an important idea: *structure improvements should matter more when fidelity is already high.*

That idea is good, but the raw formula has several problems.

3.1 Singularity at $F = 1$

The exponent $\frac{1}{1-F}$ diverges as $F \to 1$. This creates an unbounded numerical instability exactly in the regime where the training signal needs to be most reliable.

3.2 Exploding or Vanishing Reward

Let

$$B = \alpha \frac{D^* - 1}{D_{\text{current}}} + \beta \frac{C^* - 1}{C_{\text{current}}}.$$

If $B > 1$, then $B^{1/(1-F)}$ can explode. If $0 < B < 1$, then $B^{1/(1-F)}$ can collapse toward zero extremely fast.

So the fidelity coupling is too aggressive and not well controlled.

3.3 Bad Behavior When $D^* = 1$ or $C^* = 1$

The terms $D^* - 1$ and $C^* - 1$ can become zero. That can collapse the whole base even when the current circuit is good.

3.4 Poisoned References

If D^* and C^* are taken as unconstrained "best so far" values from *all* circuits, then a low-fidelity but extremely compact circuit can make the structure target unrealistically hard. This is exactly the kind of behavior that can push the model to over-optimize structure while sacrificing fidelity.

3.5 Conclusion

The right design choice is therefore:

- keep the *exponential / logarithmic coupling*;
- replace the singular exponent by a *bounded, fidelity-dependent exponent*;
- define structure references carefully;
- separate a *stable training cost* from an optional *signed diagnostic reward.*

4 Proposed Design

4.1 Notation

Let:

- $F \in [0, 1]$ be process fidelity;
- $D \in \{1, 2, \dots\}$ be estimated circuit depth;
- $C \in \{0, 1, 2, \dots\}$ be CNOT count;
- $\alpha, \beta \geq 0$ with $\alpha + \beta = 1$ be the depth / CNOT blending weights.

We also define:

- $\varepsilon_F > 0$ as a small numerical constant for fidelity shaping;
- $\varepsilon_B > 0$ as a small numerical constant for the structure term;
- $\delta_D = \delta_C = 1$ as offsets preventing division by zero.

4.2 Reference Depth and CNOT Counts

We should *not* define D^* and C^* as the best values ever seen regardless of fidelity. Instead, they should come from a fidelity-qualified set.

Choose a reference fidelity threshold τ_{ref} and define

$$\mathcal{A}_{\tau_{\text{ref}}} = \{(F_i, D_i, C_i) \ : \ F_i \geq \tau_{\text{ref}}\}.$$

Then define

$$D^* = \begin{cases} \min\{D_i : (F_i, D_i, C_i) \in \mathcal{A}_{\tau_{\text{ref}}}\}, & \mathcal{A}_{\tau_{\text{ref}}} \neq \varnothing, \\ D_{\text{fallback}}, & \mathcal{A}_{\tau_{\text{ref}}} = \varnothing, \end{cases}$$

$$C^* = \begin{cases} \min\{C_i : (F_i, D_i, C_i) \in \mathcal{A}_{\tau_{\text{ref}}}\}, & \mathcal{A}_{\tau_{\text{ref}}} \neq \varnothing, \\ C_{\text{fallback}}, & \mathcal{A}_{\tau_{\text{ref}}} = \varnothing. \end{cases}$$

Good fallback choices are:

- the depth and CNOT count of the best-fidelity circuit seen so far;
- a slowly updated EMA reference;
- or the best reference values obtained during warmup.

4.3 Fidelity Shaping

We want fidelity to dominate early, and we want high-fidelity circuits to be well separated. Define

$$\phi(F) = \text{clip}(-\log(1 - F + \varepsilon_F),\, 0,\, \phi_{\max})\,.$$

Properties:

- $\phi(F)$ is small when F is poor;
- $\phi(F)$ grows rapidly as $F \to 1$;
- clipping prevents blow-up.

4.4 Structure Ratios

Define structure-efficiency ratios

$$R_D(D) = \frac{D^* + \delta_D}{D + \delta_D}, \qquad R_C(C) = \frac{C^* + \delta_C}{C + \delta_C}.$$

These satisfy:

- $R_D = 1$ when $D = D^*$;
- $R_D > 1$ when $D < D^*$;
- $R_D < 1$ when $D > D^*$;
- and similarly for R_C.

Blend them into one structure score:

$$B(D, C) = \alpha R_D(D) + \beta R_C(C).$$

Thus:

- $B = 1$ means the circuit matches the reference structure;
- $B > 1$ means it is structurally better than the reference;
- $B < 1$ means it is structurally worse.

4.5 Bounded Fidelity-Dependent Exponent

To keep the "structure matters more as fidelity improves" idea, define a bounded exponent:

$$\kappa(F) = \kappa_{\min} + (\kappa_{\max} - \kappa_{\min}) \left[\mathrm{clip}\left(\frac{F - F_{\text{start}}}{F_{\text{full}} - F_{\text{start}}}, 0, 1 \right) \right]^p .$$

Interpretation:

- for $F \le F_{\text{start}}$, $\kappa(F) \approx \kappa_{\min}$;
- for $F \ge F_{\text{full}}$, $\kappa(F) \approx \kappa_{\max}$;
- $p > 1$ makes the ramp delayed and sharper.

This preserves the exponential idea without using the unstable factor $\frac{1}{1-F}$.

4.6 Absolute Training Score

Define the absolute circuit score

$$S(F, D, C) = (\phi(F) + \varepsilon_B)\ (B(D, C) + \varepsilon_B)^{\kappa(F)} .$$

Equivalently, in log-space,

$$s(F, D, C) = \log(\phi(F) + \varepsilon_B) + \kappa(F) \log(B(D, C) + \varepsilon_B) .$$

The log form is usually the numerically safer representation.

4.7 Training Cost

For replay-based GRPO training, use a cost rather than a signed improvement reward:

$$C_{\text{train}}(F, D, C) = -\, s(F, D, C).$$

Lower cost is better.
This is the key recommendation for the actual training loop:

- use a *stationary or slowly moving absolute cost* for replay;
- do not use raw epoch-to-epoch reward differences as the replay target.

4.8 Optional Signed Diagnostic Reward

For analysis, logging, or acceptance rules, define a signed reward relative to a reference circuit:

$$r_{\text{diag}}(F, D, C) = \big[\phi(F) - \phi(F_{\text{ref}})\big] + \mathbf{1}[F \ge F_{\text{ref}} - \Delta_F]\, \eta\, \kappa(F) \log(B(D, C) + \varepsilon_B) .$$

Here:

- $(F_{\text{ref}}, D_{\text{ref}}, C_{\text{ref}})$ is a chosen anchor circuit;
- $\Delta_F \ge 0$ is a fidelity-drop tolerance;
- $\eta > 0$ scales the structure contribution in the diagnostic reward.

This diagnostic reward has the behavior many users expect intuitively:

- if fidelity improves and structure is similar, reward is positive;
- if fidelity is unchanged and structure improves, reward is positive;
- if fidelity decreases by more than Δ_F, the structure term is suppressed and reward becomes negative.

5 Recommended Interpretation

The proposed design therefore uses *two related but distinct quantities*:

1. **Absolute training cost** C_{train} for replay-based optimization.
2. **Signed diagnostic reward** r_{diag} for interpretation and policy analysis.

This separation is important. The signed reward is easy to reason about, but the absolute cost is more suitable for stable replay training.

6 Edge-Case Analysis

Case	Behavior of the proposed design
$F = 0$	$\phi(F) \approx 0$, so $S(F, D, C)$ is near zero and C_{train} is large. Even very good structure cannot rescue a useless circuit. This is desirable.
$0 < F < F_{\text{start}}$	The exponent $\kappa(F)$ stays close to $\kappa_{\min}$, so structure matters only weakly. Training is fidelity-first.
$F = F_{\text{start}}$	This is where structure pressure begins to ramp. The reward does not jump discontinuously; it changes smoothly.
$F_{\text{start}} < F < F_{\text{full}}$	This is the trade-off region. Structure gains increasingly matter, but fidelity still has strong influence through $\phi(F)$.
$F \to 1$	$\phi(F)$ becomes large but clipped, and $\kappa(F) \to \kappa_{\max}$. High-fidelity circuits are then separated strongly by depth and CNOT count. There is no singularity.
$F = 1$ exactly	The design remains finite because $\phi(F)$ uses ε_F and clipping. Unlike $1/(1-F)$, nothing diverges.
$D = D^*, C = C^*$	Then $R_D = R_C = 1$ and $B = 1$. The structure term is neutral. The score reduces to a fidelity-only shaped score.
$D < D^*$, $C = C^*$	Then $R_D > 1$, so $B > 1$ and the score increases. If the circuit is kept, the reference depth should eventually be updated.
$D = D^*$, $C < C^*$	Then $R_C > 1$, so $B > 1$ and the score increases.
$D < D^*$ and $C < C^*$	Both structure ratios exceed one, so the structure multiplier is strongly favorable.
$D > D^*$ and $C > C^*$	Both structure ratios are below one, so the structure multiplier penalizes the circuit.
Fidelity decreases slightly, structure improves slightly	For the *training cost*, the answer depends on the full trade-off. For the *diagnostic reward*, if the fidelity drop exceeds Δ_F, the reward is forced negative. If the drop is within tolerance, the structure gain may still partially compensate.
Fidelity improves, structure worsens slightly	Early in training this is usually still favorable because fidelity dominates. Late in training the sign can flip if the structure regression is too large.
No fidelity-qualified reference exists yet	Use fallback references from warmup, best-fidelity-so-far, or a slowly moving EMA. This prevents the structure term from being defined by low-fidelity compact circuits.
$D^* = 1$ or $C^* = 0$	The $+1$ offsets keep R_D and R_C well defined. The reward does not collapse to zero.

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Case	Behavior of the proposed design
$C = 0$	Safe because of the offset $\delta_C = 1$. If zero-CNOT circuits are physically plausible for the target, they are rewarded appropriately.
Current circuit equals the reference in fidelity but has better structure	The absolute training score improves through $B > 1$. The diagnostic reward is positive because the structure term is positive while the fidelity difference term is zero.
Current circuit has lower fidelity than the reference but much better structure	This is the dangerous case. The design handles it by keeping $\kappa(F)$ small before high-fidelity regimes and by optionally suppressing the structure term in r_{diag} when fidelity drops by more than Δ_F.

7 Why This Design Fits the Project

This proposal matches the project better than a raw exponential in $\frac{1}{1-F}$ for several reasons:

- it preserves the key intuition that structure should matter more at high fidelity;
- it uses bounded, controlled nonlinearities instead of singular ones;
- it is compatible with replay-based GRPO training through an absolute cost;
- it naturally supports Pareto-style archive logic and high-fidelity structure comparisons;
- it avoids rewarding low-fidelity but compact circuits too strongly.

8 Recommended Hyperparameter Ranges

Reasonable starting ranges are:

- $\varepsilon_F = 10^{-8}$;
- $\phi_{\max} \in [10, 14]$;
- $\kappa_{\min} \in [0.05, 0.20]$;
- $\kappa_{\max} \in [1.5, 3.0]$;
- $F_{\text{start}} \in [0.90, 0.97]$;
- $F_{\text{full}} \in [0.99, 0.999]$;
- $p \in [4, 8]$;
- $\Delta_F \in [10^{-4}, 10^{-3}]$;
- α, β chosen so that the more important hardware bottleneck receives larger weight.

For three-qubit synthesis in a native `rz/sx/cx` basis, it is usually reasonable to set $\beta > \alpha$ because CNOT count often dominates hardware cost.

9 Final Recommendation

For this repository, the most defensible design is:

1. use a fidelity-shaped term $\phi(F)$;
2. use structure ratios relative to *fidelity-qualified* best-so-far references;
3. use a bounded, delayed fidelity-dependent exponent $\kappa(F)$;
4. train with the absolute cost

$$C_{\text{train}}(F, D, C) = -\log(\phi(F) + \varepsilon_B) - \kappa(F)\log(B(D, C) + \varepsilon_B);$$

5. optionally monitor the signed diagnostic reward

$$r_{\text{diag}}(F, D, C) = \big[\phi(F) - \phi(F_{\text{ref}})\big] + \mathbf{1}[F \geq F_{\text{ref}} - \Delta_F]\eta\kappa(F)\log(B(D, C) + \varepsilon_B).$$

This keeps the desirable exponential / logarithmic structure, but removes the instability of the naive $1/(1 - F)$ exponent.